

YINTONG SHANG

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SUMMARY

I am a PhD student in computer graphics, advised by Prof. Yin Yang at the University of Utah since Fall 2022. My research focuses on the intersection of computer vision and graphics, specifically on 3D reconstruction and physics-based content generation. I aspire to pursue a career in the animation and gaming industry.

EDUCATION

University of Utah 2022.8-Present
PhD of Computing, Graphics and Visualization Track, GPA: 3.9/4.0, Advisor: Prof. Yin Yang

University of Science and Technology of China (USTC) 2018.9-2022.6
Bachelor of Engineering in Electronic Science and Technology, GPA: 3.61/4.3

PROJECTS

Knowledge-driven Neural Network for Content Generation [project page](#) 2024.3-2024.6
◦ Generating accurate and coherent elastodynamics leveraging physics-in-to-loop.

Dynamic Fluid Synthesis with Gaussian Splatting [project page](#) 2023.12-2024.4
◦ Integrate physics-based animations of solids and fluids with 3D Gaussian Splatting.

Physics-based NeRF Scene Deformation [project page](#) 2023.5-2023.11
◦ Realize physics-based meshless elastic deformation to manipulate the neural radiance fields.

SKILLS

◇ Programming: C++, Python, PyTorch, CUDA, Warp Language, C#, MATLAB.
◇ Computer Graphics (CG): physics-based simulation (PBS), Newton's method, Projective Dynamics (PD), numerical optimization, collision handling, soft/rigid body simulation, cloth simulation
◇ Computer Vision (CV): 3D reconstruction, differentiable rendering, neural implicit representations (NeRF, Gaussian Splatting), parametric human body and face model, generative AI.
◇ Machine Learning (ML): neural networks (NN), deep learning.

PUBLICATIONS

* for equal contribution.

ElastoGen: 4D Generative Elastodynamics [arXiv](#) 2024.5
Yutao Feng*, Yintong Shang*, Xiang Feng*, Lei Lan, Shandian Zhe, Tianjia Shao, Hongzhi Wu, Kun Zhou, Hao Su, Chenfanfu Jiang, Yin Yang

Gaussian Splashing: Dynamic Fluid Synthesis with Gaussian Splatting [arXiv](#) 2024.1
Yutao Feng*, Xiang Feng*, Yintong Shang, Ying Jiang, Chang Yu, Zeshun Zong, Tianjia Shao, Hongzhi Wu, Kun Zhou, Chenfanfu Jiang, Yin Yang

PIE-NeRF: Physics-based Interactive Elastodynamics with NeRF (CVPR 2024) [paper](#) 2023.11
Yutao Feng*, Yintong Shang*, Xuan Li, Tianjia Shao, Chenfanfu Jiang, Yin Yang

HONORS

◇ Outstanding Student Scholarship of USTC (Silver, **Top 13%**) 2021.10
◇ The first prize in Electromagnetics Course Paper Competition (**Top 1%**) 2019.12
◇ 2018 Excellent New Student Award (**Top 20%**) 2018.12
◇ Talent Student Gold Scholarship of USTC (**Top 3%**) 2018.12
◇ Talent Program in Information Science and Technology (**Top 10%**) 2018.9-2022.6